

JKBOSE

2026 CLASS 11TH

MOTION IN PLANE

2D motion

LECTURE-2

PHYSICS

WAJAHID SIR (Gabbar



Topics To Be Covered

- 1** Revision ✓
- 2** Numerical on Projectile ✓
- 3** Uniform Circular Motion ✓
- 4** Acceleration in uniform circular motion ✓
- 5** Subtraction of two vector ✓

Main kisi se behtar banu kya
fark padta hai...,
Main kisi ka behtar karu use
se bahut fark padta hai...

Recap from 1st Lecture

- Motion in 1D



- Motion in 2D



- Motion in 3D



- Displacement in 2D

$$\vec{r} = \vec{x} + \vec{y}$$

- Velocity in 2D

$$\vec{v} = \vec{v}_x + \vec{v}_y$$

- Acceleration in 2D

$$\vec{a} = \vec{a}_x + \vec{a}_y$$

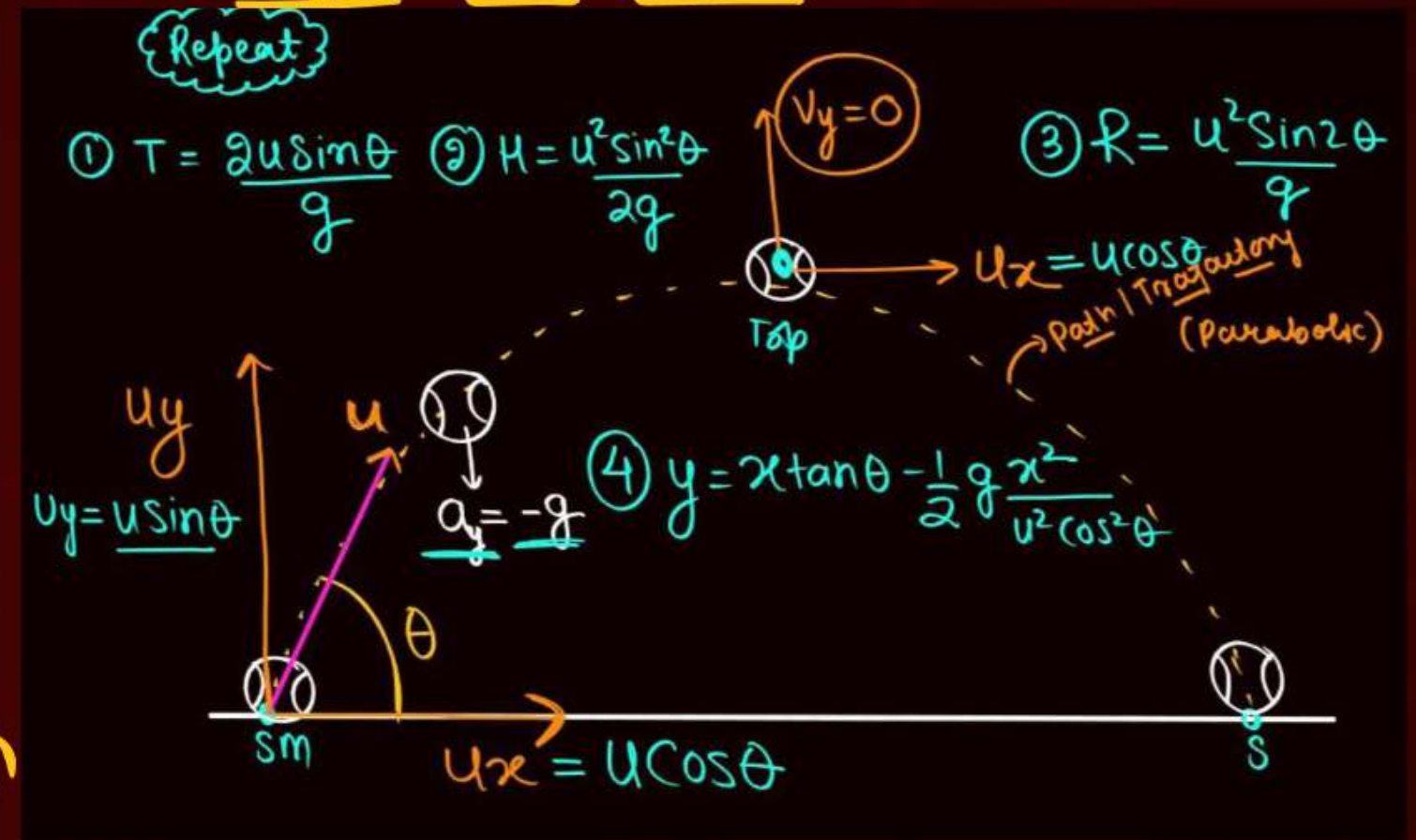
- Equation of motion

$$v = u + at \quad s = ut + \frac{1}{2}at^2 \quad v^2 = u^2 + 2as$$

$$v_x = u_x + a_x t \quad s_x = u_x t + \frac{1}{2}a_x t^2 \quad v_x^2 = u_x^2 + 2a_x s_x$$

$$v_y = u_y + a_y t \quad s_y = u_y t + \frac{1}{2}a_y t^2 \quad v_y^2 = u_y^2 + 2a_y s_y$$

Projectile Motion



Projectile Motion

(5-mark) 2026, 2025, 2024, 2023, 2022



- Example of 2D
- To explain the 2D motion
- when an object is thrown with Some Angle under the influence of gravity

Neet level ✓
 $a_y = -g = -9.8 = -10 \text{ m/s}^2$
 $a_x = 0$

2024 ✓ JKBOSE
 $v_y = 0$
Path | Tangential → Parabolic

④ $y = x \tan \theta - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$

Top $v_y = 0$ ✓

$a_y = -g$

$a = -g$

u

θ

S_m

S

Instagram Snapchat

① $T = \frac{2u \sin \theta}{g}$ ② $H = \frac{u^2 \sin^2 \theta}{2g}$ ③ $R = \frac{u^2 \sin 2\theta}{g}$

Path | Tangential → Parabolic

Projectile

① TIME PERIOD

(T)

Time of flight

The time for which object remain in Air

$$\begin{aligned} \# a_y &= -g \\ \# a_x &= 0 \end{aligned}$$

$$\begin{aligned} \# t_u &= t_d \\ \# T &= \underline{t_u} + t_d \end{aligned}$$

$$T = t_u + t_u$$

$$T = \underline{2t_u} \text{---} \textcircled{1}$$

$$T = \underline{\underline{\frac{2u \sin \theta}{g}}}$$

$$\# V = U + at$$

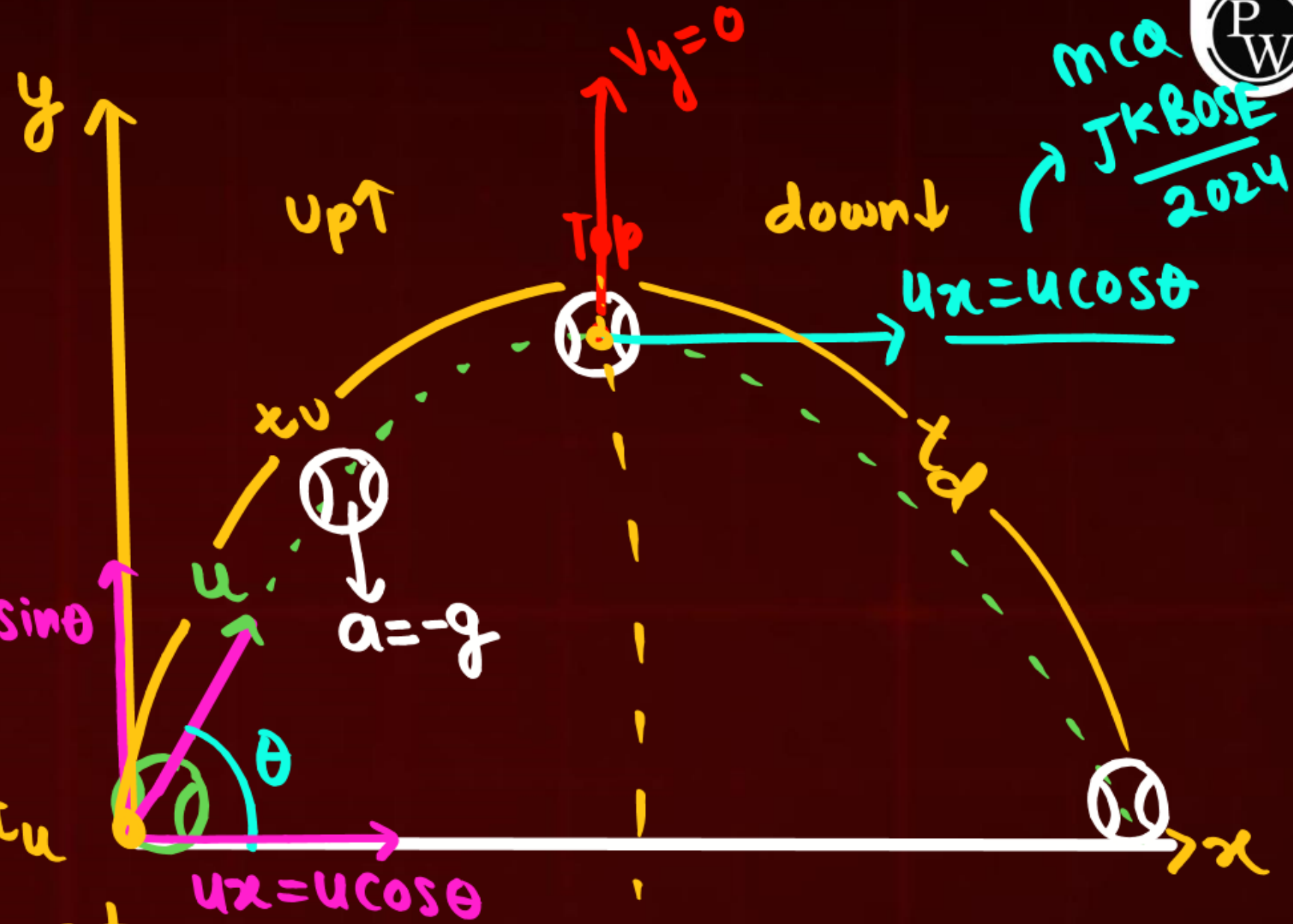
$$V_y = U_y + a_y t_u$$

$$0 = u \sin \theta + -g t_u$$

$$g t_u = u \sin \theta$$

$$t_u = \frac{u \sin \theta}{g}$$

(Put the Value of t_u in eq-①)



Projectile

HEIGHT

→ $H_{\max}$

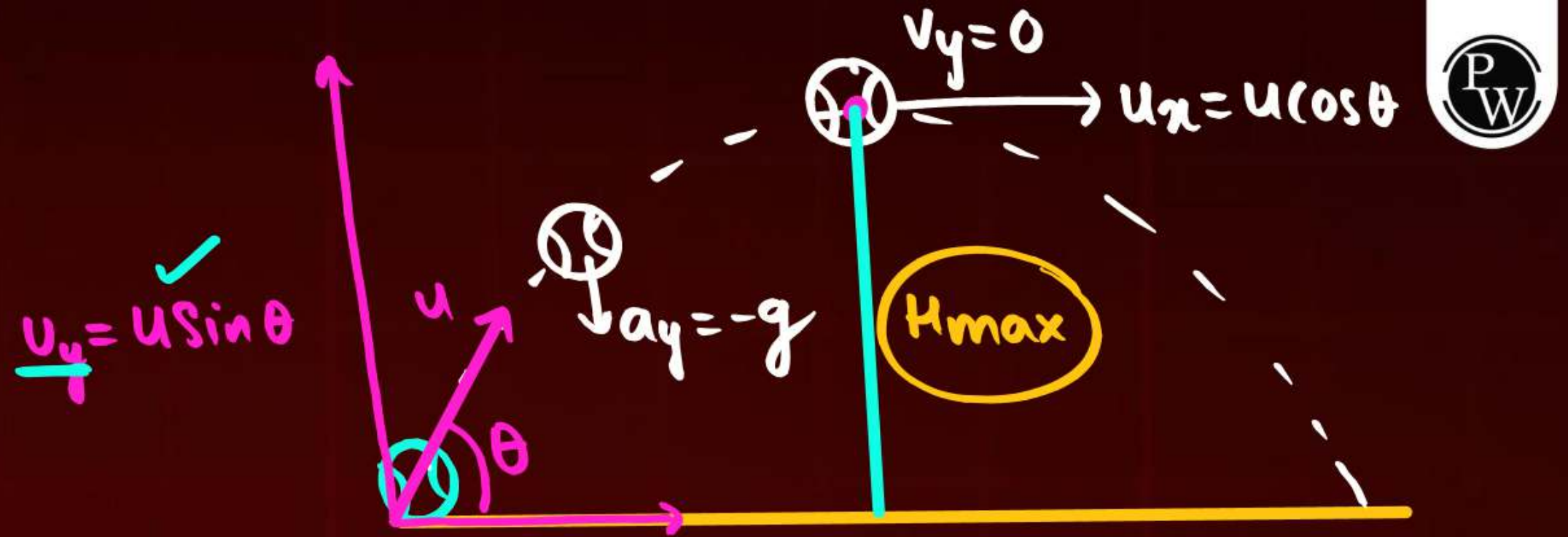
- Maximum Height
- The maximum height attain By Projectile

3rd equation

$$v^2 = u^2 + 2as$$

$$v_y^2 = u_y^2 + 2a_y H_{\max}$$

$$0^2 = (u \sin \theta)^2 + 2(-g) H_{\max}$$



$$0 = u^2 \sin^2 \theta - 2g H_{\max}$$

$$2g H_{\max} = u^2 \sin^2 \theta$$

$$H_{\max} = \frac{u^2 \sin^2 \theta}{2g}$$

$$(u \sin \theta)^2 \rightarrow u^2 \sin^2 \theta$$

Projectile

RANGE

→ x-displacement

The Horizontal displacement
of the object

$$\# \quad \underline{v} = \frac{d}{t}$$

$$\underline{u_x} = \frac{x}{T}$$

$$u \cos \theta T = x$$

$$R/x = u \cos \theta \quad \textcircled{T}$$

$$R = u \cos \theta \frac{2u \sin \theta}{g}$$

$$R = \frac{u^2 2 \sin \theta \cos \theta}{g}$$

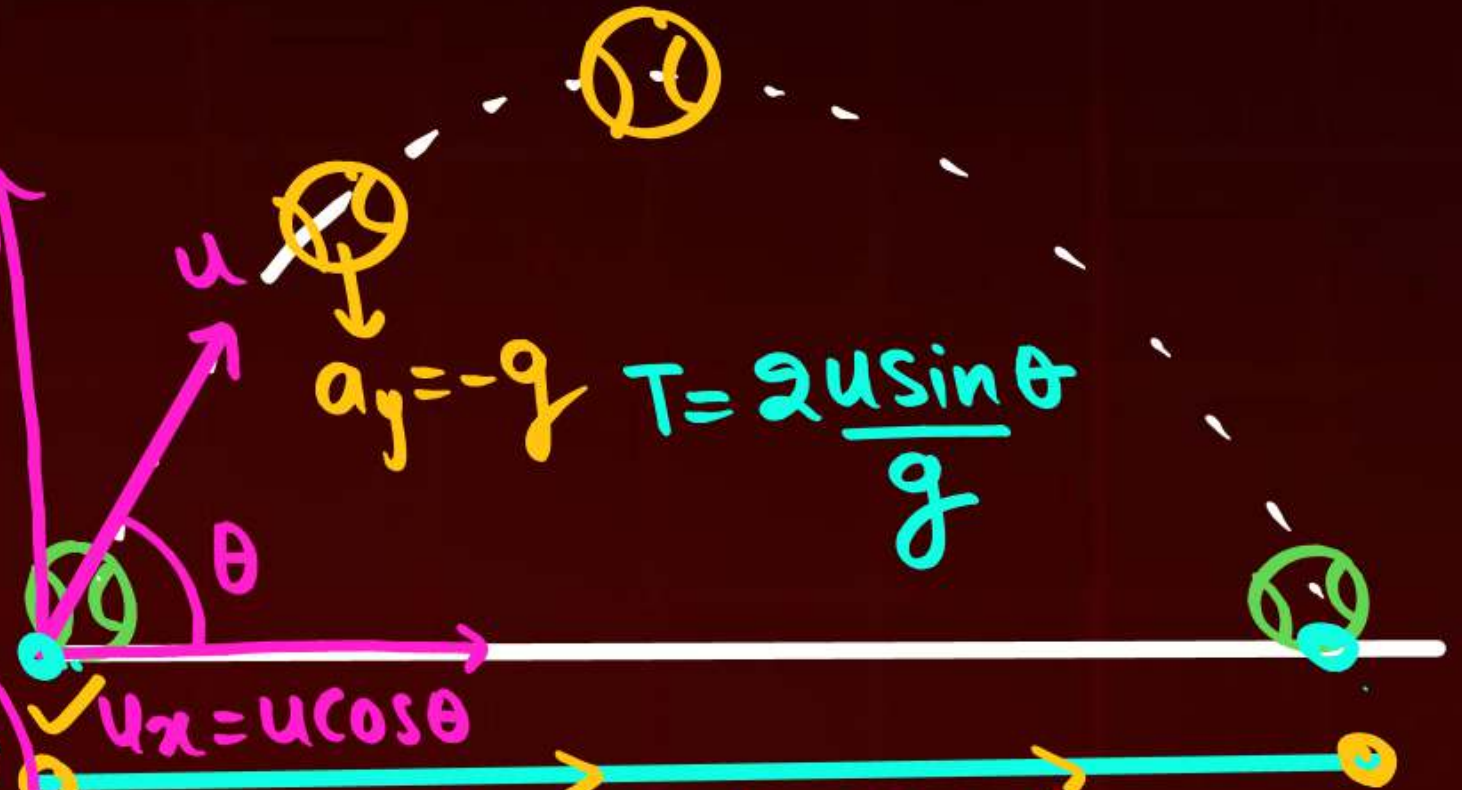
$$R = \frac{u^2 \sin 2\theta}{g} \quad \checkmark$$

$$\underline{2 \sin \theta \cos \theta = \sin 2\theta}$$

$\times u_y = u \sin \theta$
 u
 $a_y = -g$
 $T = \frac{2u \sin \theta}{g}$

$u_x = u \cos \theta$

x-Displacement
Range

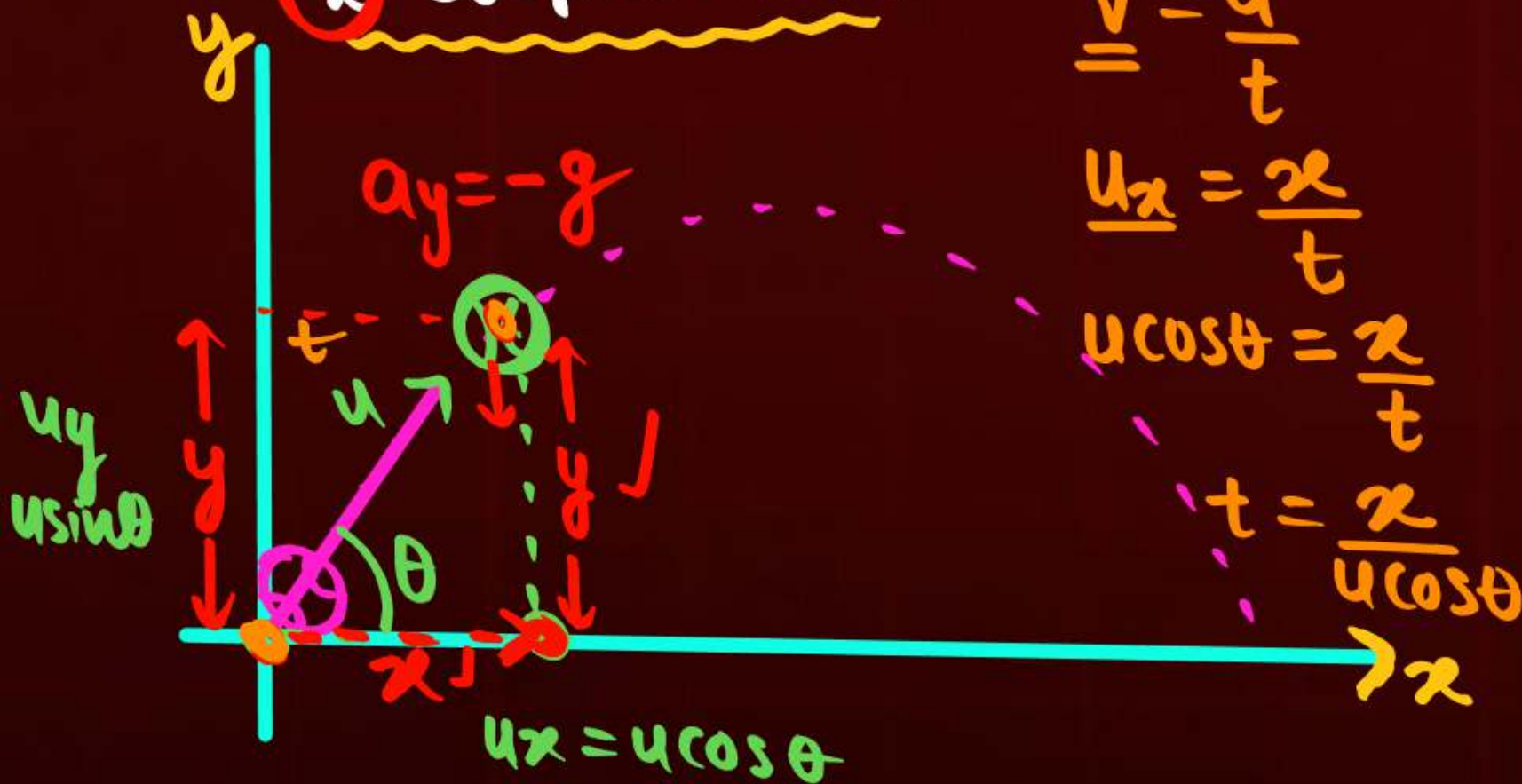


Projectile

2025

④ EQUATION OF TRAJECTORY (5m)

- The equation which tell us about the Relationship b/w y-displacement and x-displacement



$$v = \frac{d}{t}$$

$$u \cos \theta = \frac{x}{t}$$

$$t = \frac{x}{u \cos \theta}$$

2nd equation of Motion ✓

$$S_y = ut + \frac{1}{2}at^2$$

$$y = u \sin \theta t + \frac{1}{2}(-g)t^2$$

$$y = u \sin \theta t - \frac{1}{2}gt^2 \quad (\text{Relationship})$$

$y \rightarrow t$

$$y = u \sin \theta \frac{x}{u \cos \theta} - \frac{1}{2}g \left(\frac{x}{u \cos \theta} \right)^2$$

$$y = \frac{\sin \theta}{\cos \theta} x - \frac{1}{2}g \frac{x^2}{u^2 \cos^2 \theta}$$

$$y = \tan \theta x - \frac{1}{2}g \frac{x^2}{u^2 \cos^2 \theta}$$

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2024

$$\bullet y = \tan \theta x - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$$

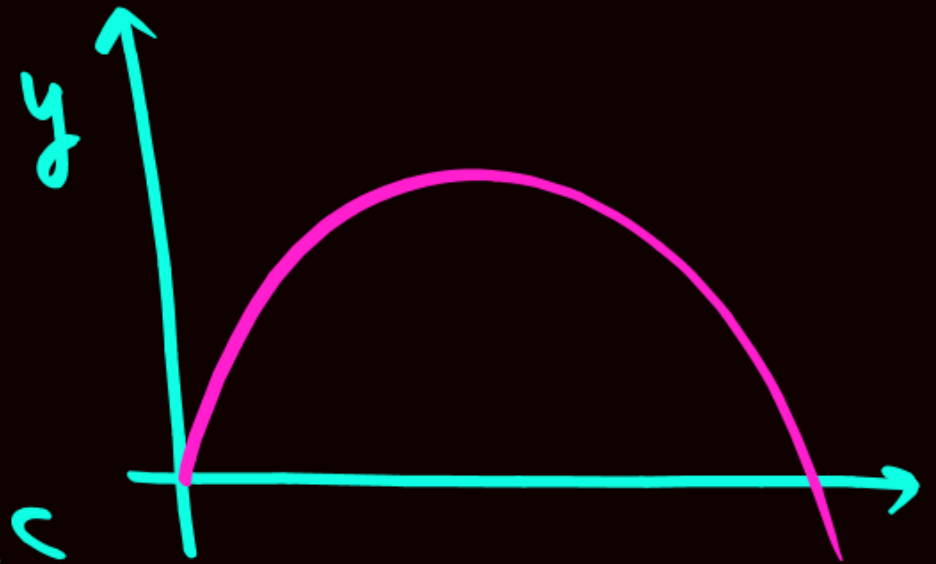
$$\underline{y} = \underline{x} \tan \theta - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$$

Proof
Trajectory Parabolic

$$y = \underline{x} \frac{\tan \theta}{A} - \frac{1}{2} \frac{g}{u^2 \cos^2 \theta} x^2$$

$$y = xA - Bx^2$$

→ Parabolic



Chocolate Question

A at which angle the range is max JKBOSE (2019)



$$R = \frac{u^2 \sin 2\theta}{g}$$

$$R_{\text{max}} = \frac{u^2}{g} \sin 2\theta_{\text{max}}$$

$$\sin 2\theta = 1$$

$$\sin 2\theta = \sin 90^\circ$$

$$2\theta = 90^\circ$$

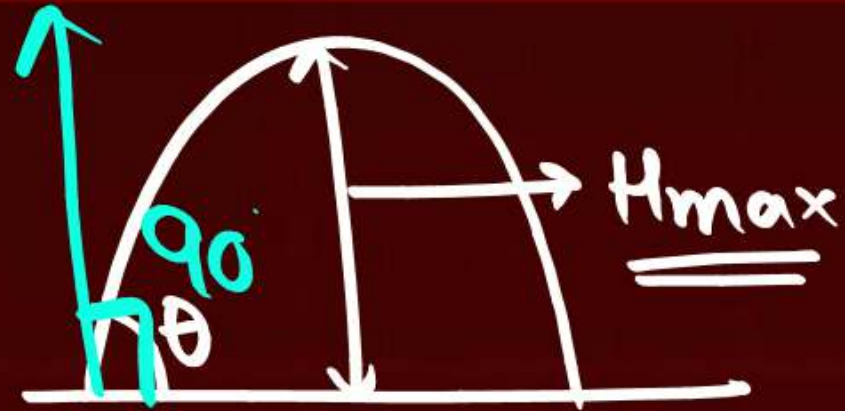
$$\theta = \frac{90}{2} = 45^\circ$$

- A 30°
- B 120°
- C 90°
- D 45° ✓

Chocolate Question

At which angle the height of projectile is max

- A** 45°
- B** 90° ✓
- C** 120°
- D** 150°



$$H_{\text{max}} = \frac{u^2 \sin^2 \theta}{2g}$$

max

$$\sin^2 \theta = 1$$

$$\sin \theta = \sqrt{1}$$

$$\sin \theta = 1$$

$$\sin \theta = \sin 90^\circ$$

$$\theta = 90^\circ$$

Chocolate Question

At which angle the ^{Time} period of projectile is max

H/W

- A** 45°
- B** 60°
- C** 70°
- D** 90°

SUMMARY



Projectile

T

H

R

of

2

Eid Mubarrak

Eididi



Thank
You